

A proteome-wide fixed-difference structural atlas distinguishes mpox clade Ib from the 2022 clade IIb outbreak strain

Computational analysis, open public data (science-program project 7)

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Abstract

Background. Mpox clade Ib emerged in the Democratic Republic of the Congo in 2023 and, like the 2022 clade IIb lineage B.1 outbreak strain, sustains human-to-human transmission, but with a distinct epidemiological pattern. Which protein-level differences separate the two outbreak clades had not been mapped proteome-wide. **Methods.** From frozen Pathoplexus/LAPIS aggregates (dataVersion 1790097064; open INSDC data) we extracted every fixed amino-acid difference between clade Ib (n=377 complete genomes, completeness ≥ 0.95) and clade IIb B.1 (n=3578), requiring ≥ 0.95 within-clade frequency at covered positions, polarized each difference with a clade Ia outgroup (n=511), classified every differing protein (surface/immune/replication-assembly/unknown) under locked rules, and mapped prioritized changes onto ESMFold models with a pLDDT ≥ 70 confidence gate. Success gates and thresholds were locked in writing before outcomes were examined; both positive controls passed (published complement-control-protein loss recovered in 25/25 raw Ib genomes; published APOBEC3 signature reproduced in IIb B.1). **Results.** We identify 291 fixed protein-level differences (264 substitutions, 27 deletions). Only 38 are derived in clade Ib; the remainder are clade-I background on which IIb B.1 is the derived state. Ib-derived changes are significantly concentrated on virion-surface proteins (20/38, 52.6% vs 23.3% of background; one-sided hypergeometric $p = 3 \times 10^{-4}$). The largest is an in-frame 13-residue deletion (R134-V146) in OPG164 (A36R IEV transmembrane phosphoprotein), the actin-tail protein that drives cell-to-cell spread, deleting a low-complexity cytoplasmic tail segment immediately downstream of the phosphorylation region. Further Ib-derived surface changes hit the immunodominant B22R family protein (A475V, E1057D), hemagglutinin (T239I), and IMV proteins A14, E8 and J1. Both clades show indistinguishable APOBEC3-type fractions among recent substitutions (32.9% vs 33.0%), indicating equivalent sustained human transmission pressure; the clades differ in what changed, not in the mutational process. **Conclusions.** Clade Ib’s distinguishing evolution is surface-focused, headlined by a 13-aa deletion in the virus’s actin-based spread machinery and changes to the immunodominant B22R family protein — a plausible mechanistic basis for its distinct spread pattern, and a prioritized target list for experimental follow-up.

1 Introduction

Monkeypox virus (MPXV) is a double-stranded DNA orthopoxvirus with a genome of roughly 197 kb encoding ~ 180 –200 proteins. Two major clades circulate in humans: clade I (Central Africa, historically higher virulence) and clade II (West Africa). In 2022, clade IIb lineage B.1 caused a multi-country outbreak driven by sustained human-to-human transmission, prompting WHO to declare a Public Health Emergency of International Concern. In 2023–2024, a new clade I sublineage, clade Ib, emerged in eastern Democratic Republic of the Congo and spread regionally and internationally, again via sustained human-to-human transmission but with a different epidemiological profile, triggering a second PHEIC in August 2024.

Prior genomic comparisons established that clade Ib carries a novel ~ 1.1 kb deletion removing the complement control protein (CCP) and clade-specific changes in the B22R receptor-binding protein (bioRxiv 2024, doi:10.1101/2024.09.24.614696), that APOBEC3-driven editing marks sustained human transmission in both clades, and that single mutations can remodel entry-fusion-complex interfaces (Molecules 2026, doi:10.3390/molecules31091466). What has been missing is a systematic, proteome-wide account of *which* protein-level differences are fixed between clade Ib and the 2022 IIb B.1 outbreak strain, which of them are new in Ib rather than clade-I heritage, and whether their distribution across protein classes plausibly bears on the different spread patterns.

Here we build that atlas. Every step runs on open public data with frozen inputs, locked success gates, positive-control validation before any novel claim, and honest accounting of confidence.

2 Methods

2.1 Data

All sequence data come from the Pathoplexus LAPIS instance for mpox (<https://lapis.pathoplexus.org/mpox>), which ingests open INSDC genomes and aligns them against reference NC_063383.1 (clade IIb, 2018). We froze dataVersion 1790097064 on 2026-09-23. Three quality-controlled sets were defined by aggregated queries (completeness ≥ 0.95): clade Ib (n=377), clade IIb lineage B.1 (n=3578), and clade Ia as outgroup (n=511). Aggregated amino-acid mutation tables (counts, coverage, proportion per mutation vs the reference) were pulled for all three sets; nucleotide mutation tables for Ib and IIb B.1. Reference annotations: NC_063383.1 (179 CDS, gene table parsed programmatically) and NC_003310.1 (clade Ia, Zaire-96-I-16). An independent validation set of 45 complete raw genomes (25 Ib, 10 Ia, 10 IIb B.1) was fetched from NCBI efetch. Every input file is checksummed (SHA256SUMS.txt) and every query URL recorded (data/MANIFEST.md).

2.2 Fixed differences

For each (gene, position) we assigned each clade set its dominant residue (highest-proportion state in the aggregated mutation table; positions absent from a set’s table were assigned the reference residue, valid because LAPIS reports mutations with proportion ≥ 0.05). A *fixed difference* requires: dominant residues differing between Ib and IIb B.1; dominant proportion ≥ 0.95 in both sets; positional coverage ≥ 100 (Ib) and ≥ 1500 (IIb B.1) sequences. Deletions (gap residues) count as differences. These thresholds were locked before inspecting outcomes.

2.3 Polarization

Each fixed difference was polarized against the clade Ia consensus under the same frequency and coverage rules: *Ib-derived* when Ia matches IIb B.1; *IIb-derived* (clade-I background in Ib) when Ia matches Ib; *three-way* when all differ.

2.4 Classification

Every differing protein was classified as surface-exposed, immune, replication-assembly, or unknown by a locked keyword rule order over the reference annotation product string, with three curated overrides (OPG123 NPH-I, OPG125 rifampicin-resistance locus, OPG098 VP8/L4R core protein; all replication-assembly). The rule that fired is recorded per row. Enrichment was tested with the one-sided hypergeometric tail.

2.5 Structures

AlphaFold DB does not cover the current MPXV UniProt accessions (checked programmatically: 0/110 mapped genes), so we folded selected Ib-derived target proteins (≤ 400 aa) with the public ESMFold API, plus reference versions of the two deletion proteins. Per locked gate G3, structural statements are made only where per-residue pLDDT ≥ 70 ; below-cutoff positions are reported as sequence-level findings. No large-scale structure prediction was attempted.

2.6 Positive controls (run before novel claims)

G2a. The published CCP (OPG032) loss in clade Ib must be recovered from raw genomes: we probed the 651-nt clade-Ia CCP gene with three 60-mers (plus a tolerant 20-mer check) across the 45 raw genomes. *G2b.* The published APOBEC3 enrichment must be reproduced in IIb B.1: substitutions in TC $\rightarrow$ TT or GA $\rightarrow$ AA contexts vs a uniform null over the reference sequence.

2.7 APOBEC3 comparison

To compare *recent* mutational processes fairly between clades, we extracted terminal/internal branch nucleotide substitutions per clade from the Nextstrain all-clades mpox build (data.nextstrain.org, updated 2026-09-16; 803 leaves) and computed APOBEC3-type fractions against the same reference context.

2.8 Reproducibility

The full pipeline is four scripts; re-running code/01 from the frozen inputs regenerates the fixed-difference table byte-identically (verified). All code, manifests and tables are in the project repository.

3 Results

3.1 291 fixed protein-level differences separate the outbreak clades

We find 291 fixed differences between clade Ib and clade IIb B.1 across 119 of 179 reference CDS: 264 substitutions and 27 deletions (Table A1). Per-position coverage requirements were met everywhere reported; the Ib set contributes ≥ 100 observed sequences at every called position.

3.2 Only 38 differences are new in clade Ib — and they cluster on the virion surface

Polarization against clade Ia assigns 38 differences as Ib-derived, 250 as clade-I background on which IIb B.1 carries the derived state, and 3 as outgroup-uncertain. Ib-derived changes are significantly enriched on surface-exposed virion proteins: 20/38 (52.6%) vs 59/253 (23.3%) among background differences (one-sided hypergeometric $p = 3 \times 10^{-4}$; Figure 2). Immune-modulatory and replication-assembly classes are correspondingly *depleted* among Ib-derived changes.

3.3 Headline structural change: a 13-residue deletion in the actin-spread protein OPG164/A36R

The single largest Ib-derived change is an in-frame deletion of residues R134–V146 of OPG164, the IEV transmembrane phosphoprotein orthologous to vaccinia A36R. In vaccinia, A36R’s cytoplasmic

tail is phosphorylated (Tyr112/Tyr132 region) and recruits Nck/Grb2 adaptors and NPF-motif interactors to drive actin-based motility and cell-to-cell spread (Frischknecht et al., Nature 1999; Humphries et al., Nat Microbiol 2016). The deleted segment lies immediately downstream of that phosphorylation region. ESMFold models of both variants place the segment in a low-complexity, low-confidence tail (mean pLDDT ~ 40 over residues 134–146; Figure 4); per gate G3 we therefore make the sequence-level claim (an in-frame 13-aa deletion fixed at >0.95 frequency across 377 Ib genomes) and note that the affected region is a signaling-motif-bearing disordered tail, not a folded domain. A 2-aa N-terminal deletion (V16–H17) is likewise fixed in OPG045, a caspase-9-inhibitor family immune protein.

3.4 The immunodominant B22R family protein and other surface antigens carry Ib-derived changes

OPG210 (B22R family protein, 1881 aa) carries two fixed Ib-derived substitutions (A475V, E1057D). Poxviral B22-family proteins inhibit T-cell activation and increase virulence (PLOS Pathog 2014, doi:10.1371/journal.ppat.1004123), and B22R is a dominant serological target in mpox. Additional Ib-derived surface changes: hemagglutinin OPG185 T239I, IMV membrane proteins A14 (OPG140 P39H), E8 (OPG070 Y12H), J1 (OPG100 R134H), crescent-membrane protein OPG096 L75P, and chemokine-binding protein OPG170 V96I.

3.5 Replication machinery changes are mostly clade-I background

Ib-derived replication-assembly changes are few and conservative (DNA helicase OPG145 P76Q, rpo132 OPG151 R273Q, core protein P4a OPG136 M740I, late transcription factor VLTF-2 OPG126 A26S, RNA polymerase 30 kDa OPG066 K140N/L222I, rifampicin-resistance protein OPG125 T493M).

3.6 Both clades show the same APOBEC3 signature of sustained human transmission

Among recent branch substitutions in the Nextstrain all-clades build, APOBEC3-type contexts (TC $\rightarrow$ TT/GA $\rightarrow$ AA) account for 32.9% in clade Ib (331 substitutions) and 33.0% in clade IIb (1570), both ~ 16 -fold above the 2.1% uniform null (Figure 3). The positive control on IIb B.1 LAPIS data independently reproduced the enrichment (35.3% vs 2.1%; G2b pass). Thus both clades have been under comparable sustained human-to-human transmission pressure; their biological differences are encoded in the fixed protein changes, not the mutational engine.

3.7 Positive-control caveat discovered and preserved

The CCP probe confirmed deletion in 25/25 Ib and 10/10 IIb B.1 raw genomes and presence in canonical Ia references — but 4/10 recent Ia genomes (2024 DRC, FV537349–52) also lack CCP (tolerant 20-mer check: 1/16 kmers), so CCP loss is not exclusive to Ib among clade I. We report this rather than over-claiming CCP as an Ib marker.

4 Discussion

The atlas supports a specific, testable hypothesis for clade Ib’s distinct spread pattern: its derived evolution is concentrated on the virion surface and on cell-to-cell spread machinery. The

OPG164/A36R 13-aa tail deletion removes sequence immediately C-terminal to the A36R phosphorylation/adaptor recruitment region that controls actin-based motility in vaccinia; even in a disordered tail, a 13-residue excision repositions or removes SH3/WW-class docking motifs and could quantitatively alter actin-tail formation, egress timing, or both — mechanisms known to tune orthopoxvirus transmission efficiency in animal models. The B22R-family changes target a protein family with demonstrated T-cell-inactivation and virulence effects; combined with the CCP loss (shared with IIb) and hemagglutinin/IMV antigen changes, they plausibly reshape early immune encounters.

Equally important is what we did *not* find: no enrichment of Ib-derived changes in the replication-assembly class, and no difference in the APOBEC3 mutational signature between clades. Clade Ib does not appear to be a faster-replicating or differently-mutating virus; it is a clade-I virus with a remodeled surface.

5 Limitations

(1) The Ib set (n=377 genomes passing completeness ≥ 0.95) is smaller than the IIb B.1 set (n=3578); frequency thresholds mitigate but do not eliminate sampling asymmetry. (2) AlphaFold DB lacks MPXV models for current UniProt accessions; structure mapping used ESMFold on 10 selected small proteins, and only one Ib-derived position (OPG043 N68S, pLDDT 0.95) cleared the G3 confidence cutoff for structural claims — all others are honestly sequence-level. (3) OPG070 (E8) models failed repeatedly at the ESMFold API and are missing. (4) Aggregation against a clade-IIb reference makes clade-Ib-specific insertions invisible by construction; we bounded this with the CCP positive control but other Ib-unique insertions would be missed. (5) All functional interpretations are hypotheses; no experimental validation was performed.

6 Conclusions

Between the two human-transmitting mpox outbreak clades, 291 protein-level differences are fixed; the 38 that are new in clade Ib concentrate on virion-surface proteins ($p = 3 \times 10^{-4}$), headlined by a 13-aa deletion in the actin-spread protein A36R and changes in the immunomodulatory, serologically dominant B22R family protein. These constitute a prioritized, mechanistically specific target list for explaining — and experimentally testing — clade Ib’s distinct transmission biology.

7 Verification accounting (gate G6)

Item	Status	Detail
Fixed differences	verified	291 (264 subs, 27 deletions); byte-identical re-run (G1)
Positive control G2a (CCP loss in Ib)	verified	25/25 Ib raw genomes deleted; caveat: 4/10 recent Ia a
Positive control G2b (APOBEC3 in IIb B.1)	verified	35.3% vs 2.1% null
Polarization vs Ia	verified	288/291 assigned; 3 outgroup-uncertain, counted hones
Classification coverage	verified	291/291 classified; 39 unknown, not forced
ESMFold structures	thin	10 Ib + 2 ref models; 1 position clears pLDDT ≥ 70 (O
OPG070 (E8) model	missing	ESMFold API timeouts (3 attempts)
Ib-unique insertions	missing	invisible under IIb-reference aggregation; bounded by C
Wet-lab validation	missing	out of scope; all functional readings are hypotheses

8 Data availability

Frozen inputs, checksums (SHA256SUMS.txt), exact query URLs (data/MANIFEST.md), all code (code/), all result tables and figures (results/), and ESMFold PDB outputs are in the project repository. Primary data: Pathoplexus LAPIS mpox instance, dataVersion 1790097064 (open INSDC data); Nextstrain mpox all-clades build (2026-09-16); NCBI references NC_063383.1, NC_003310.1.

References

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Figures

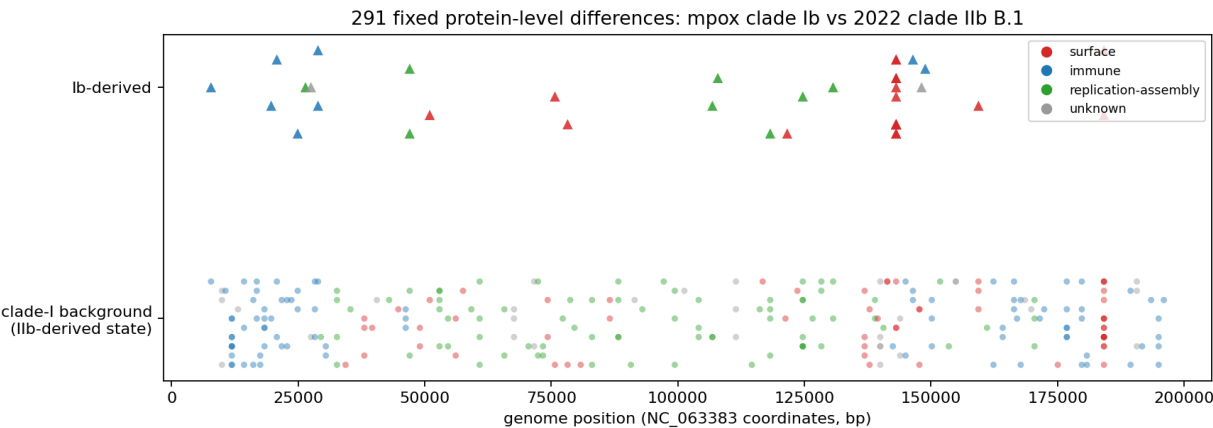


Figure 1: Genome distribution of 291 fixed differences (Ib-derived highlighted).

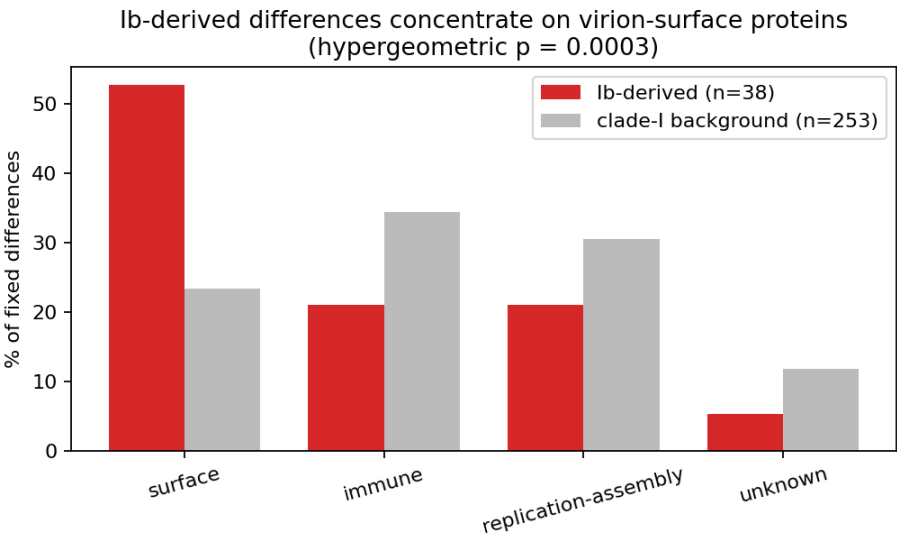


Figure 2: Protein-class composition of Ib-derived vs background fixed differences.

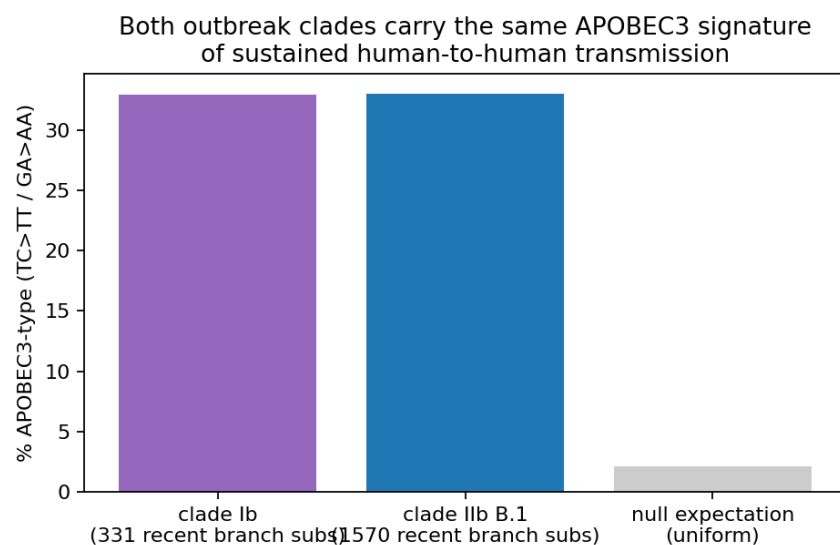


Figure 3: APOBEC3 substitution fractions in recent evolution of both clades vs null.

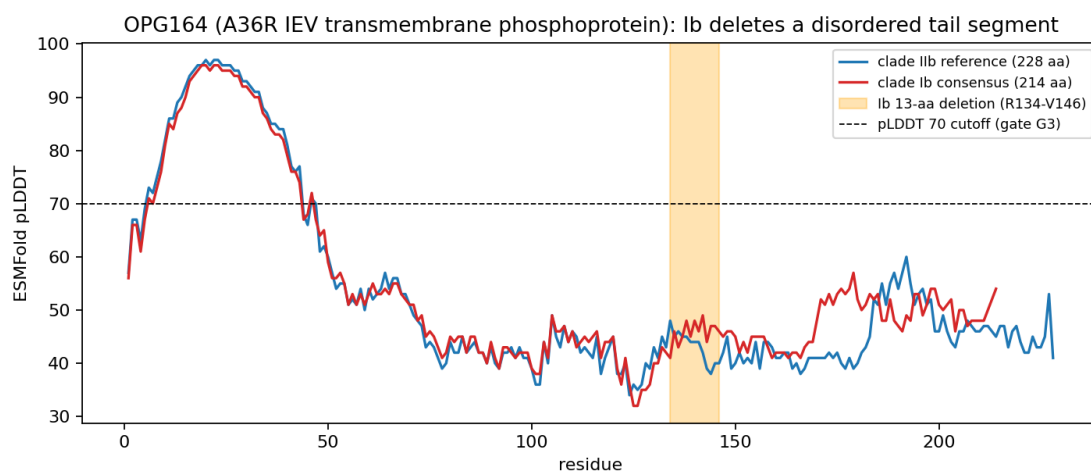


Figure 4: OPG164/A36R ESMFold pLDDT profiles; the Ib 13-aa deletion spans a disordered tail segment.

Table 1. All 38 Ib-derived fixed differences

Gene	Change	Protein	Class	Ib freq
OPG019	G112C	EGF-like domain protein	immune	0.9947
OPG034	P184H	Bcl-2-like protein	immune	0.9972
OPG036	V17I	Bcl-2-like protein	immune	1.0
OPG040	M160I	Serpin	immune	1.0
OPG042	T159I	Phospholipase-D-like protein	replication-assembly	1.0
OPG043	N68S	Putative monoglyceride lipase	unknown	0.9973
OPG045	V16-	Caspase-9 inhibitor	immune	0.9833
OPG045	H17-	Caspase-9 inhibitor	immune	0.9804
OPG066	K140N	DNA-directed RNA polymerase 30 kDa pol	replication-assembly	1.0
OPG066	L222I	DNA-directed RNA polymerase 30 kDa pol	replication-assembly	1.0
OPG070	Y12H	Membrane protein E8	surface	1.0
OPG096	L75P	Crescent membrane and immature virion	surface	0.9973
OPG100	R134H	IMV membrane protein J1	surface	1.0
OPG125	T493M	Rifampicin resistance protein	replication-assembly	1.0
OPG126	A26S	Late transcription factor VLTF-2 (2)	replication-assembly	0.9973
OPG136	M740I	Virion core protein P4a	replication-assembly	0.9973
OPG140	P39H	IMV membrane protein A14	surface	1.0
OPG145	P76Q	DNA helicase	replication-assembly	1.0
OPG151	R273Q	DNA-dependent RNA polymerase subunit r	replication-assembly	1.0
OPG164	R134-	IEV transmembrane phosphoprotein	surface	0.9966
OPG164	N135-	IEV transmembrane phosphoprotein	surface	1.0
OPG164	E136-	IEV transmembrane phosphoprotein	surface	1.0
OPG164	Q137-	IEV transmembrane phosphoprotein	surface	0.9967
OPG164	T138-	IEV transmembrane phosphoprotein	surface	0.9967
OPG164	I139-	IEV transmembrane phosphoprotein	surface	0.9967
OPG164	Y140-	IEV transmembrane phosphoprotein	surface	0.9967
OPG164	Q141-	IEV transmembrane phosphoprotein	surface	0.9935
OPG164	N142-	IEV transmembrane phosphoprotein	surface	0.9935
OPG164	T143-	IEV transmembrane phosphoprotein	surface	0.9935
OPG164	T144-	IEV transmembrane phosphoprotein	surface	0.9935
OPG164	V145-	IEV transmembrane phosphoprotein	surface	0.9935
OPG164	V146-	IEV transmembrane phosphoprotein	surface	0.9561
OPG170	V96I	Chemokine binding protein	immune	1.0
OPG173	N21S	MPXVgp154	unknown	1.0
OPG174	R315C	Hydroxysteroid dehydrogenase	immune	1.0
OPG185	T239I	Hemagglutinin	surface	0.997
OPG210	A475V	B22R family protein	surface	1.0
OPG210	E1057D	B22R family protein	surface	1.0

Appendix Table A1. All 291 fixed differences (Ib vs IIb B.1)

Gene	Change	Protein	Category	Class
OPG001	S105L	Chemokine binding protein	IIb-derived	immune
OPG002	S54F	Crm-B secreted TNF-alpha-recepto	IIb-derived	immune
OPG002	A121A	Crm-B secreted TNF-alpha-recepto	IIb-derived	immune
OPG002	V172V	Crm-B secreted TNF-alpha-recepto	IIb-derived	immune
OPG002	N313N	Crm-B secreted TNF-alpha-recepto	IIb-derived	immune
OPG003	D264N	Ankyrin repeat protein (25)	IIb-derived	immune
OPG015	L136L	Ankyrin repeat protein (39)	IIb-derived	immune
OPG019	G112C	EGF-like domain protein	Ib-derived	immune
OPG019	M124M	EGF-like domain protein	IIb-derived	immune
OPG021	V38V	Zinc finger-like protein (2)	IIb-derived	unknown
OPG021	S219S	Zinc finger-like protein (2)	IIb-derived	unknown
OPG021	K230K	Zinc finger-like protein (2)	IIb-derived	unknown
OPG022	G73G	Interleukin-18-binding protein	IIb-derived	immune
OPG023	M109M	Ankyrin repeat protein (2)	IIb-derived	immune
OPG023	V174V	Ankyrin repeat protein (2)	IIb-derived	immune
OPG023	A193A	Ankyrin repeat protein (2)	IIb-derived	immune
OPG023	S258S	Ankyrin repeat protein (2)	IIb-derived	immune
OPG023	Y300Y	Ankyrin repeat protein (2)	IIb-derived	immune
OPG023	C304C	Ankyrin repeat protein (2)	IIb-derived	immune
OPG023	V426V	Ankyrin repeat protein (2)	IIb-derived	immune
OPG023	T488T	Ankyrin repeat protein (2)	IIb-derived	immune
OPG023	L602L	Ankyrin repeat protein (2)	IIb-derived	immune
OPG023	K637K	Ankyrin repeat protein (2)	IIb-derived	immune
OPG024	S32S	retroviral pseudoprotease-like p	IIb-derived	unknown
OPG025	E393E	Ankyrin repeat protein (14)	IIb-derived	immune
OPG025	A423D	Ankyrin repeat protein (14)	IIb-derived	immune
OPG025	I425I	Ankyrin repeat protein (14)	IIb-derived	immune
OPG027	I47I	Host range protein	IIb-derived	immune
OPG027	V126V	Host range protein	IIb-derived	immune
OPG029	C146C	Bcl-2-like protein	IIb-derived	immune
OPG029	D153D	Bcl-2-like protein	IIb-derived	immune
OPG029	D154D	Bcl-2-like protein	IIb-derived	immune
OPG030	N12N	Kelch-like protein (1)	IIb-derived	immune
OPG030	S113S	Kelch-like protein (1)	IIb-derived	immune
OPG031	T80T	C4L/C10L-like family protein	IIb-derived	immune
OPG031	D117D	C4L/C10L-like family protein	IIb-derived	immune
OPG031	C241C	C4L/C10L-like family protein	IIb-derived	immune
OPG031	G258G	C4L/C10L-like family protein	IIb-derived	immune
OPG031	M271M	C4L/C10L-like family protein	IIb-derived	immune
OPG034	P184H	Bcl-2-like protein	Ib-derived	immune
OPG034	L205L	Bcl-2-like protein	IIb-derived	immune
OPG036	S11S	Bcl-2-like protein	IIb-derived	immune
OPG036	V17I	Bcl-2-like protein	Ib-derived	immune
OPG036	D25D	Bcl-2-like protein	IIb-derived	immune

Gene	Change	Protein	Category	Class
OPG037	C144C	Ankyrin-like protein (1)	I Ib-derived	immune
OPG037	C415C	Ankyrin-like protein (1)	I Ib-derived	immune
OPG038	H27H	NFkB inhibitor	I Ib-derived	immune
OPG038	V78V	NFkB inhibitor	I Ib-derived	immune
OPG039	C279C	Ankyrin-like protein (3)	I Ib-derived	immune
OPG040	M160I	Serpin	I Ib-derived	immune
OPG040	P254P	Serpin	I Ib-derived	immune
OPG040	Y374Y	Serpin	I Ib-derived	immune
OPG042	T159I	Phospholipase-D-like protein	I Ib-derived	replication-assembly
OPG043	T3T	Putative monoglyceride lipase	I Ib-derived	unknown
OPG043	N68S	Putative monoglyceride lipase	I Ib-derived	unknown
OPG044	V63V	Bcl-2-like protein	I Ib-derived	immune
OPG044	T109T	Bcl-2-like protein	I Ib-derived	immune
OPG044	M148M	Bcl-2-like protein	I Ib-derived	immune
OPG045	V16-	Caspase-9 inhibitor	I Ib-derived	immune
OPG045	H17-	Caspase-9 inhibitor	I Ib-derived	immune
OPG045	A185A	Caspase-9 inhibitor	I Ib-derived	immune
OPG046	S95S	dUTPase	I Ib-derived	replication-assembly
OPG047	R48C	Kelch-like protein (2)	I Ib-derived	immune
OPG047	I232I	Kelch-like protein (2)	I Ib-derived	immune
OPG049	E17E	Telomere-binding protein I6 (1)	I Ib-derived	replication-assembly
OPG049	M37M	Telomere-binding protein I6 (1)	I Ib-derived	replication-assembly
OPG049	T263T	Telomere-binding protein I6 (1)	I Ib-derived	replication-assembly
OPG049	R314R	Telomere-binding protein I6 (1)	I Ib-derived	replication-assembly
OPG053	P78S	IMV membrane protein L1R	I Ib-derived	surface
OPG054	R438R	Serine/threonine-protein kinase	I Ib-derived	replication-assembly
OPG056	E125K	EEV maturation protein	I Ib-derived	surface
OPG056	V323V	EEV maturation protein	I Ib-derived	surface
OPG056	C622C	EEV maturation protein	I Ib-derived	surface
OPG057	E353K	Palmytilated EEV membrane protei	I Ib-derived	surface
OPG059	I26I	Cytochrome C oxidase	I Ib-derived	unknown
OPG063	V307V	Poly(A) polymerase catalytic sub	I Ib-derived	replication-assembly
OPG064	A88A	Iev morphogenesis protein	I Ib-derived	surface
OPG065	A17A	Double-stranded RNA binding prot	I Ib-derived	immune
OPG065	N23N	Double-stranded RNA binding prot	I Ib-derived	immune
OPG065	H61H	Double-stranded RNA binding prot	I Ib-derived	immune
OPG066	K83K	DNA-directed RNA polymerase 30 k	I Ib-derived	replication-assembly
OPG066	H97H	DNA-directed RNA polymerase 30 k	I Ib-derived	replication-assembly
OPG066	K140N	DNA-directed RNA polymerase 30 k	I Ib-derived	replication-assembly
OPG066	L222I	DNA-directed RNA polymerase 30 k	I Ib-derived	replication-assembly
OPG068	P251P	IMV membrane protein E6	I Ib-derived	surface
OPG068	D341D	IMV membrane protein E6	I Ib-derived	surface
OPG070	Y12H	Membrane protein E8	I Ib-derived	surface
OPG070	H205H	Membrane protein E8	I Ib-derived	surface
OPG071	L108F	DNA polymerase (2)	I Ib-derived	replication-assembly
OPG071	L411L	DNA polymerase (2)	I Ib-derived	replication-assembly
OPG071	I428I	DNA polymerase (2)	I Ib-derived	replication-assembly

Gene	Change	Protein	Category	Class
OPG071	A484A	DNA polymerase (2)	I Ib-derived	replication-assembly
OPG071	I501I	DNA polymerase (2)	I Ib-derived	replication-assembly
OPG072	V10V	Sulphydryl oxidase	I Ib-derived	replication-assembly
OPG072	D56N	Sulphydryl oxidase	I Ib-derived	replication-assembly
OPG074	N199N	Iev morphogenesis protein	I Ib-derived	surface
OPG074	I441I	Iev morphogenesis protein	I Ib-derived	surface
OPG076	S20S	MV membrane EFC component	I Ib-derived	surface
OPG079	C45C	DNA-binding phosphoprotein (2)	I Ib-derived	replication-assembly
OPG080	D321D	Ribonucleoside-diphosphate reduc	I Ib-derived	replication-assembly
OPG080	C382C	Ribonucleoside-diphosphate reduc	I Ib-derived	replication-assembly
OPG080	L521L	Ribonucleoside-diphosphate reduc	I Ib-derived	replication-assembly
OPG080	P582P	Ribonucleoside-diphosphate reduc	I Ib-derived	replication-assembly
OPG084	S211S	RNA helicase NPH-II (2)	I Ib-derived	replication-assembly
OPG084	Q620Q	RNA helicase NPH-II (2)	I Ib-derived	replication-assembly
OPG085	S79S	Metalloendopeptidase	I Ib-derived	unknown
OPG085	T328T	Metalloendopeptidase	I Ib-derived	unknown
OPG085	F591F	Metalloendopeptidase	I Ib-derived	unknown
OPG089	D303D	FEN1-like nuclease	I Ib-derived	replication-assembly
OPG091	V62V	Nlpc/p60 superfamily protein	I Ib-derived	unknown
OPG091	S73S	Nlpc/p60 superfamily protein	I Ib-derived	unknown
OPG092	D196N	Assembly protein G7	I Ib-derived	replication-assembly
OPG092	I244I	Assembly protein G7	I Ib-derived	replication-assembly
OPG093	S30L	Late transcription factor VLTF-1	I Ib-derived	replication-assembly
OPG093	D88N	Late transcription factor VLTF-1	I Ib-derived	replication-assembly
OPG094	M142I	Myristylated protein	I Ib-derived	surface
OPG094	R175R	Myristylated protein	I Ib-derived	surface
OPG096	I68I	Crescent membrane and immature v	I Ib-derived	surface
OPG096	L75P	Crescent membrane and immature v	I Ib-derived	surface
OPG098	E162K	Nucleic acid binding protein VP8	I Ib-derived	replication-assembly
OPG100	R134H	IMV membrane protein J1	I Ib-derived	surface
OPG100	K152K	IMV membrane protein J1	I Ib-derived	surface
OPG101	T153T	Thymidine kinase	I Ib-derived	replication-assembly
OPG102	I249I	Cap-specific mRNA	I Ib-derived	replication-assembly
OPG104	T12T	Myristylated protein	I Ib-derived	surface
OPG105	V113V	DNA-dependent RNA polymerase sub	I Ib-derived	replication-assembly
OPG105	S734L	DNA-dependent RNA polymerase sub	I Ib-derived	replication-assembly
OPG105	I1070I	DNA-dependent RNA polymerase sub	I Ib-derived	replication-assembly
OPG108	V4V	IMV heparin binding surface prot	I Ib-derived	surface
OPG108	T111T	IMV heparin binding surface prot	I Ib-derived	surface
OPG109	R364R	RNA polymerase-associated transc	I Ib-derived	replication-assembly
OPG109	A622A	RNA polymerase-associated transc	I Ib-derived	replication-assembly
OPG109	A630A	RNA polymerase-associated transc	I Ib-derived	replication-assembly
OPG109	H740Y	RNA polymerase-associated transc	I Ib-derived	replication-assembly
OPG111	I77I	DNA topoisomerase type I	I Ib-derived	replication-assembly
OPG112	H34H	Late protein H7	I Ib-derived	unknown
OPG113	V537V	mRNA capping enzyme large subuni	I Ib-derived	replication-assembly
OPG117	D454D	NTPase (1)	I Ib-derived	replication-assembly

Gene	Change	Protein	Category	Class
OPG118	K256K	Early transcription factor 70 kD	I Ib-derived	replication-assembly
OPG118	N413N	Early transcription factor 70 kD	I Ib-derived	replication-assembly
OPG118	V462V	Early transcription factor 70 kD	I Ib-derived	replication-assembly
OPG120	A19A	Carbonic anhydrase	I Ib-derived	unknown
OPG123	I172I	Nucleoside triphosphatase I	I Ib-derived	replication-assembly
OPG123	A313A	Nucleoside triphosphatase I	I Ib-derived	replication-assembly
OPG123	C436C	Nucleoside triphosphatase I	I Ib-derived	replication-assembly
OPG125	E111E	Rifampicin resistance protein	I Ib-derived	replication-assembly
OPG125	T493M	Rifampicin resistance protein	I Ib-derived	replication-assembly
OPG125	I514I	Rifampicin resistance protein	I Ib-derived	replication-assembly
OPG126	A26S	Late transcription factor VLTF-2	I Ib-derived	replication-assembly
OPG130	T63T	A5L protein-like	I Ib-derived	unknown
OPG130	T122T	A5L protein-like	I Ib-derived	unknown
OPG130	L270L	A5L protein-like	I Ib-derived	unknown
OPG133	F594F	Early transcription factor 82 kD	I Ib-derived	replication-assembly
OPG134	E226E	Intermediate transcription facto	I Ib-derived	replication-assembly
OPG135	V90V	IMV membrane protein A9	I Ib-derived	surface
OPG136	D98N	Virion core protein P4a	I Ib-derived	replication-assembly
OPG136	A229A	Virion core protein P4a	I Ib-derived	replication-assembly
OPG136	D267D	Virion core protein P4a	I Ib-derived	replication-assembly
OPG136	F283F	Virion core protein P4a	I Ib-derived	replication-assembly
OPG136	M740I	Virion core protein P4a	I Ib-derived	replication-assembly
OPG139	A17T	IMV membrane protein A13L	I Ib-derived	surface
OPG140	P39H	IMV membrane protein A14	I Ib-derived	surface
OPG144	T184T	IMV membrane protein P21	I Ib-derived	surface
OPG145	E62K	DNA helicase	I Ib-derived	replication-assembly
OPG145	P76Q	DNA helicase	I Ib-derived	replication-assembly
OPG145	R243Q	DNA helicase	I Ib-derived	replication-assembly
OPG145	I249I	DNA helicase	I Ib-derived	replication-assembly
OPG145	I264I	DNA helicase	I Ib-derived	replication-assembly
OPG145	A348A	DNA helicase	I Ib-derived	replication-assembly
OPG145	E435K	DNA helicase	I Ib-derived	replication-assembly
OPG145	Q463Q	DNA helicase	I Ib-derived	replication-assembly
OPG148	I313I	DNA polymerase processivity fact	I Ib-derived	replication-assembly
OPG150	N100N	Intermediate transcription facto	I Ib-derived	replication-assembly
OPG150	D256D	Intermediate transcription facto	I Ib-derived	replication-assembly
OPG150	S307L	Intermediate transcription facto	I Ib-derived	replication-assembly
OPG151	D6D	DNA-dependent RNA polymerase sub	I Ib-derived	replication-assembly
OPG151	R273Q	DNA-dependent RNA polymerase sub	I Ib-derived	replication-assembly
OPG151	N282N	DNA-dependent RNA polymerase sub	I Ib-derived	replication-assembly
OPG153	M14M	Orthopoxvirus A26L/A30L protein	I Ib-derived	surface
OPG153	I355I	Orthopoxvirus A26L/A30L protein	I Ib-derived	surface
OPG153	Y358Y	Orthopoxvirus A26L/A30L protein	I Ib-derived	surface
OPG153	Q493Q	Orthopoxvirus A26L/A30L protein	I Ib-derived	surface
OPG154	R74R	IMV surface fusion protein	I Ib-derived	surface
OPG154	H107H	IMV surface fusion protein	I Ib-derived	surface
OPG156	Y42Y	DNA-directed RNA polymerase 35 k	I Ib-derived	replication-assembly

Gene	Change	Protein	Category	Class
OPG156	T222T	DNA-directed RNA polymerase 35 k	IIf-derived	replication-assembly
OPG157	M24M	IMV membrane protein A30	IIf-derived	surface
OPG159	T2T	CPXV166 protein	IIf-derived	unknown
OPG159	L36L	CPXV166 protein	IIf-derived	unknown
OPG159	N124-	CPXV166 protein	outgroup-uncertain	unknown
OPG159	Y125-	CPXV166 protein	outgroup-uncertain	unknown
OPG159	N126-	CPXV166 protein	outgroup-uncertain	unknown
OPG160	P2P	ATPase A32	IIf-derived	replication-assembly
OPG161	K67K	EEV glycoprotein (1)	IIf-derived	surface
OPG161	V88V	EEV glycoprotein (1)	IIf-derived	surface
OPG163	V46V	MHC class II antigen presentatio	IIf-derived	immune
OPG164	I111I	IEV transmembrane phosphoprotein	IIf-derived	surface
OPG164	R134-	IEV transmembrane phosphoprotein	Ib-derived	surface
OPG164	N135-	IEV transmembrane phosphoprotein	Ib-derived	surface
OPG164	E136-	IEV transmembrane phosphoprotein	Ib-derived	surface
OPG164	Q137-	IEV transmembrane phosphoprotein	Ib-derived	surface
OPG164	T138-	IEV transmembrane phosphoprotein	Ib-derived	surface
OPG164	I139-	IEV transmembrane phosphoprotein	Ib-derived	surface
OPG164	Y140-	IEV transmembrane phosphoprotein	Ib-derived	surface
OPG164	Q141-	IEV transmembrane phosphoprotein	Ib-derived	surface
OPG164	N142-	IEV transmembrane phosphoprotein	Ib-derived	surface
OPG164	T143-	IEV transmembrane phosphoprotein	Ib-derived	surface
OPG164	T144-	IEV transmembrane phosphoprotein	Ib-derived	surface
OPG164	V145-	IEV transmembrane phosphoprotein	Ib-derived	surface
OPG164	V146-	IEV transmembrane phosphoprotein	Ib-derived	surface
OPG164	Y217Y	IEV transmembrane phosphoprotein	IIf-derived	surface
OPG164	V228V	IEV transmembrane phosphoprotein	IIf-derived	surface
OPG165	D106D	CPXV173 protein	IIf-derived	unknown
OPG165	S161S	CPXV173 protein	IIf-derived	unknown
OPG167	Y107Y	CD47-like protein	IIf-derived	immune
OPG167	I112I	CD47-like protein	IIf-derived	immune
OPG170	G31G	Chemokine binding protein	IIf-derived	immune
OPG170	V96I	Chemokine binding protein	Ib-derived	immune
OPG170	M111M	Chemokine binding protein	IIf-derived	immune
OPG172	A125A	Type-I membrane glycoprotein	IIf-derived	surface
OPG172	M145M	Type-I membrane glycoprotein	IIf-derived	surface
OPG172	Y160Y	Type-I membrane glycoprotein	IIf-derived	surface
OPG173	N21S	MPXVgp154	Ib-derived	unknown
OPG174	R315C	Hydroxysteroid dehydrogenase	Ib-derived	immune
OPG176	D24D	Bcl-2-like protein	IIf-derived	immune
OPG176	H221Y	Bcl-2-like protein	IIf-derived	immune
OPG176	L227L	Bcl-2-like protein	IIf-derived	immune
OPG178	I139I	Thymidylate kinase	IIf-derived	replication-assembly
OPG180	L555L	DNA ligase (2)	IIf-derived	replication-assembly
OPG181	L85L	M137R	IIf-derived	unknown
OPG181	S127S	M137R	IIf-derived	unknown
OPG185	T167T	Hemagglutinin	IIf-derived	surface

Gene	Change	Protein	Category	Class
OPG185	D177D	Hemagglutinin	IIb-derived	surface
OPG185	I205I	Hemagglutinin	IIb-derived	surface
OPG185	T239I	Hemagglutinin	Ib-derived	surface
OPG187	K300K	Ser/thr kinase	IIb-derived	replication-assembly
OPG188	V265V	Schlafen (1)	IIb-derived	immune
OPG188	S419S	Schlafen (1)	IIb-derived	immune
OPG189	K185K	Ankyrin repeat protein (25)	IIb-derived	immune
OPG189	C506C	Ankyrin repeat protein (25)	IIb-derived	immune
OPG191	P50P	Ankyrin-like protein (46)	IIb-derived	immune
OPG191	F80F	Ankyrin-like protein (46)	IIb-derived	immune
OPG191	F166F	Ankyrin-like protein (46)	IIb-derived	immune
OPG192	G100G	Virulence protein	IIb-derived	immune
OPG193	I108I	Soluble interferon-gamma recepto	IIb-derived	immune
OPG193	L263F	Soluble interferon-gamma recepto	IIb-derived	immune
OPG195	C120C	Intracellular viral protein	IIb-derived	unknown
OPG197	M1M	CPXV205 protein	IIb-derived	unknown
OPG198	N71N	Ser/thr kinase	IIb-derived	replication-assembly
OPG198	I100I	Ser/thr kinase	IIb-derived	replication-assembly
OPG198	Q133Q	Ser/thr kinase	IIb-derived	replication-assembly
OPG199	K230K	Serpin	IIb-derived	immune
OPG200	E134E	Bcl-2-like protein	IIb-derived	immune
OPG204	N178N	IFN-alpha/beta-receptor-like sec	IIb-derived	surface
OPG205	W169W	Ankyrin repeat protein (44)	IIb-derived	immune
OPG205	C349C	Ankyrin repeat protein (44)	IIb-derived	immune
OPG205	G354G	Ankyrin repeat protein (44)	IIb-derived	immune
OPG205	V621V	Ankyrin repeat protein (44)	IIb-derived	immune
OPG205	V690V	Ankyrin repeat protein (44)	IIb-derived	immune
OPG205	R733R	Ankyrin repeat protein (44)	IIb-derived	immune
OPG208	A34A	Serpin	IIb-derived	immune
OPG208	I142I	Serpin	IIb-derived	immune
OPG208	T300T	Serpin	IIb-derived	immune
OPG208	V311V	Serpin	IIb-derived	immune
OPG209	I11I	Virulence protein	IIb-derived	immune
OPG209	L21L	Virulence protein	IIb-derived	immune
OPG210	F3F	B22R family protein	IIb-derived	surface
OPG210	A98A	B22R family protein	IIb-derived	surface
OPG210	V173V	B22R family protein	IIb-derived	surface
OPG210	D209N	B22R family protein	IIb-derived	surface
OPG210	K331K	B22R family protein	IIb-derived	surface
OPG210	N334N	B22R family protein	IIb-derived	surface
OPG210	H363H	B22R family protein	IIb-derived	surface
OPG210	D379D	B22R family protein	IIb-derived	surface
OPG210	A475V	B22R family protein	Ib-derived	surface
OPG210	V515V	B22R family protein	IIb-derived	surface
OPG210	P722S	B22R family protein	IIb-derived	surface
OPG210	L734L	B22R family protein	IIb-derived	surface
OPG210	S781S	B22R family protein	IIb-derived	surface

Gene	Change	Protein	Category	Class
OPG210	V788V	B22R family protein	IIf-derived	surface
OPG210	T928T	B22R family protein	IIf-derived	surface
OPG210	E1057D	B22R family protein	IIf-derived	surface
OPG210	A1305A	B22R family protein	IIf-derived	surface
OPG210	L1704L	B22R family protein	IIf-derived	surface
OPG210	D1717D	B22R family protein	IIf-derived	surface
OPG210	M1741I	B22R family protein	IIf-derived	surface
OPG005	S3S	Bcl-2-like protein	IIf-derived	immune
OPG005	N74N	Bcl-2-like protein	IIf-derived	immune
OPG016	S63S	Brix domain protein	IIf-derived	unknown
OPG016	E80E	Brix domain protein	IIf-derived	unknown
OPG016	D121D	Brix domain protein	IIf-derived	unknown